

# SESSION 8 — GITHUB LAB

## Security in Networking

Security Groups • NACLs • Open Port Risks • Misconfiguration Scenarios

---

### ◆ LAB OVERVIEW

#### LAB TITLE

Enterprise AWS Networking Security Lab

#### LAB TYPE

GitHub-Based Practical Security Review

#### DURATION

45–60 Minutes

#### OBJECTIVE

Students will learn how to:

- ✓ Identify insecure networking configurations
  - ✓ Understand Security Groups and NACLs
  - ✓ Detect dangerous open ports
  - ✓ Analyze cloud security risks
  - ✓ Understand layered network protection
  - ✓ Apply enterprise security best practices
- 

### ◆ REQUIRED TOOLS

#### STUDENTS NEED

- ✓ GitHub Account
  - ✓ VS Code or Browser
  - ✓ Internet Connection
- 

## ◆ RECOMMENDED GITHUB REPOSITORY STRUCTURE

```
network-security-lab/
├── README.md
├── screenshots/
│   ├── insecure-security-group.png
│   ├── secure-security-group.png
│   └── nacl-example.png
├── scenarios/
│   ├── scenario-1-public-ssh.md
│   └── scenario-2-public-database.md
├── notes/
│   ├── sg-vs-nacl.md
│   └── open-port-risks.md
└── answers/
    └── remediation-notes.md
```

---

## ◆ SECTION 1 — SECURITY GROUP ANALYSIS

### FILE

```
notes/sg-vs-nacl.md
```

---

## ◆ INSECURE SECURITY GROUP EXAMPLE

### SCREENSHOT CONTENT

```
Type: SSH
Protocol: TCP
Port: 22
Source: 0.0.0.0/0
```

---

## ◆ STUDENT TASK

### ANALYZE THE CONFIGURATION

#### QUESTIONS

1. Which port is open?
2. Why is this risky?
3. What does 0.0.0.0/0 mean?
4. What type of attack may happen?
5. How should this be secured?

---

## ◆ EXPECTED STUDENT ANSWERS

- ✓ Port 22 is open
- ✓ SSH is publicly exposed
- ✓ Entire internet can connect
- ✓ Brute-force attacks may occur
- ✓ SSH should be restricted to trusted IPs

---

## ◆ SECTION 2 — SECURE SECURITY GROUP

### SCREENSHOT CONTENT

Type: HTTPS  
Protocol: TCP  
Port: 443  
Source: 0.0.0.0/0

Type: SSH  
Protocol: TCP  
Port: 22  
Source: Trusted-IP

---

## ◆ STUDENT TASK

## IDENTIFY WHY THIS IS BETTER

### EXPECTED ANSWERS

- ✓ HTTPS traffic is allowed securely
  - ✓ SSH access is restricted
  - ✓ Reduced attack surface
  - ✓ Better governance and security
- 

## ◆ SECTION 3 — SECURITY GROUPS VS NACLs

### FILE

`notes/sg-vs-nacl.md`

---

## ◆ COMPARISON TABLE

| Feature     | Security Groups | NACLs  |
|-------------|-----------------|--------|
| Level       | Instance        | Subnet |
| Stateful    | Yes             | No     |
| Allow Rules | Yes             | Yes    |
| Deny Rules  | No              | Yes    |
| Scope       | Resource        | Subnet |

---

## ◆ STUDENT TASK

### QUESTIONS

1. Which service protects EC2 instances?
  2. Which service protects subnets?
  3. Which one is stateful?
  4. Which one supports deny rules?
- 

## ◆ EXPECTED ANSWERS

- ✓ Security Groups protect instances
  - ✓ NACLs protect subnets
  - ✓ Security Groups are stateful
  - ✓ NACLs support deny rules
- 

## ◆ SECTION 4 — OPEN PORT RISKS

### FILE

`notes/open-port-risks.md`

---

## ◆ HIGH-RISK PORTS

| Port | Risk                |
|------|---------------------|
| 22   | SSH attacks         |
| 3389 | RDP attacks         |
| 3306 | Database exposure   |
| 5432 | PostgreSQL exposure |

---

## ◆ STUDENT TASK

### QUESTIONS

1. Why is Port 22 dangerous publicly?
  2. Why should databases remain private?
  3. What is port scanning?
  4. What happens when attackers discover open ports?
- 

## ◆ EXPECTED ANSWERS

- ✓ Attackers may attempt brute-force attacks
  - ✓ Databases contain sensitive data
  - ✓ Port scanning searches for vulnerable systems
  - ✓ Systems become attack targets
-

# ◆ SECTION 5 — MISCONFIGURATION SCENARIO 1

## FILE

scenarios/scenario-1-public-ssh.md

---

## ◆ SCENARIO

An administrator creates this rule:

Port: 22  
Source: 0.0.0.0/0

---

## ◆ STUDENT TASK

### IDENTIFY THE RISKS

#### EXPECTED ANSWERS

- ✓ SSH exposed publicly
  - ✓ Entire internet allowed
  - ✓ Brute-force attacks possible
  - ✓ Unauthorized access risk
- 

## ◆ ENTERPRISE FIXES

- ✓ Restrict SSH access
  - ✓ Use trusted IP addresses
  - ✓ Use VPN access
  - ✓ Enable MFA
- 

# ◆ SECTION 6 — MISCONFIGURATION SCENARIO 2

# FILE

scenarios/scenario-2-public-database.md

---

## ◆ SCENARIO

A company deploys a database publicly.

Database Access:  
Publicly Accessible = YES

---

## ◆ STUDENT TASK

### IDENTIFY THE RISKS

#### EXPECTED ANSWERS

- ✓ Database exposure
  - ✓ Data theft risk
  - ✓ Compliance violations
  - ✓ Possible ransomware attacks
- 

## ◆ ENTERPRISE FIXES

- ✓ Move database to private subnet
  - ✓ Restrict access
  - ✓ Enable encryption
  - ✓ Monitor traffic continuously
- 

## ◆ SECTION 7 — AWS CONSOLE REVIEW ACTIVITY

### STUDENTS SHOULD REVIEW

- ✓ Security Group rules
- ✓ Open ports

- ✓ Allowed IP ranges
  - ✓ Public vs private access
  - ✓ Network security layers
- 

## ◆ SAFE CONFIGURATION EXAMPLE

HTTPS (443) → Public Allowed  
SSH (22) → Restricted  
Database → Private Only

---

## ◆ DANGEROUS CONFIGURATION EXAMPLE

SSH (22) → Public  
RDP (3389) → Public  
Database (3306) → Public

---

## ◆ SECTION 8 — QUICK KNOWLEDGE CHECK

### QUESTION 1

Which firewall protects subnets?

- A. Security Group
- B. NACL

✓ Correct Answer: B

---

### QUESTION 2

Which firewall is stateful?

- A. Security Group
- B. NACL

✓ Correct Answer: A

---

## QUESTION 3

Which configuration is dangerous?

Port: 22  
Source: 0.0.0.0/0

✓ Correct Answer: Dangerous

---

## ◆ FINAL KEY TAKEAWAYS

- ✓ Security Groups protect instances
  - ✓ NACLs protect subnets
  - ✓ Open ports create severe risks
  - ✓ Misconfigurations cause breaches
  - ✓ Layered defense improves security
  - ✓ Governance creates resilient environments
- 

## ◆ FINAL MESSAGE

Great cloud professionals secure environments before attacks happen.

Strong cloud security begins with disciplined network protection and governance.